

Quality of Service (Part 2)

- Topics covered:
 - Classification/Marking
 - Queuing/Congestion Management
 - Shaping/Policing

Classification/Marking

Introduction to QoS Classification

QoS (Quality of Service) gives important network traffic higher priority during periods of network congestion.

QoS consists of three major functions:

1. **Classification and Marking**
 - Identifies which traffic should receive higher priority.
2. **Queuing and Congestion Management**
 - Determines how packets are stored and forwarded during congestion.
3. **Shaping and Policing**
 - Controls the rate at which traffic enters or exits an interface.

Classification is the first and most important step because a device must identify traffic before it can prioritize it.

What is Classification?

Classification is the process of organizing network traffic (packets) into different traffic classes or categories.

It is fundamental to QoS because routers and switches must first recognize which packets require special treatment before they can assign different priorities.

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Methods of Traffic Classification

Traffic can be classified in several ways.

A. Access Control List (ACL)

An ACL can identify specific traffic.

- Traffic matching (permitted by) the ACL can receive special treatment, such as high priority.
- Traffic not matching the ACL will not receive that treatment.

Here, the ACL is **not** used to permit or deny traffic. It is only used to identify specific traffic, similar to its use with Dynamic NAT.

B. NBAR (Network Based Application Recognition)

Sometimes Layer 3 and Layer 4 information alone is not enough to identify the exact application.

NBAR performs **Deep Packet Inspection (DPI)** by examining packet information beyond Layers 3 and 4 up to **Layer 7**, allowing it to identify the specific application generating the traffic.

This lesson focuses instead on the dedicated QoS fields found in packet headers.

QoS Classification Fields

QoS uses dedicated fields in Layer 2 and Layer 3 headers for traffic classification.

Layer	Field	Purpose
Layer 2	PCP (Priority Code Point)	Identifies high- or low-priority traffic in an 802.1Q VLAN tag
Layer 3	DSCP (Differentiated Services Code Point)	Identifies traffic priority in the IP header

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PCP (Priority Code Point)

PCP is located inside the **802.1Q VLAN tag**.

It is available only when an **Ethernet frame** contains an **802.1Q** (dot1q) VLAN tag.

Ethernet Header



802.1Q Tag Format



Important Points

1. PCP is a 3-bit field.
2. It is also called CoS (Class of Service).
3. CoS refers only to the PCP field, not the overall QoS concept.
4. PCP is defined by IEEE 802.1p.

PCP Values

Since PCP uses **3 bits**, it supports: $2^3 = 8$ possible values 0–7

PCP value	Traffic types
0	Best effort (default)
1	Background
2	Excellent effort
3	Critical applications
4	Video
5	Voice
6	Internetwork control
7	Network control

1. 0 → Best Effort
2. 3 → Critical Applications
3. 4 → Video
4. 5 → Voice

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Best Effort (PCP = 0)

Best effort means:

- No guarantee that data will be delivered.
- No guarantee that QoS requirements will be met.
- Represents normal, non-priority traffic.
- Default PCP value for regular traffic.

Voice Traffic PCP Values

IP phones use different PCP values for different types of voice traffic.

PCP Traffic

- 3 Call signaling traffic (used to establish phone calls)
- 5 Actual voice traffic (audio packets)

IP phones **mark** call signaling traffic (used to establish calls) as PCP3.
They **mark** the actual voice traffic as PCP5.

Marking

Marking is the process of assigning a value to either:

- the PCP field, or
- the DSCP field.

Routers and switches examine these markings and classify packets according to their priority.

For example:

- An IP phone marks audio packets with **PCP 5**.
- Network devices recognize that marking and classify those packets as high-priority traffic.

Where PCP Can Be Used

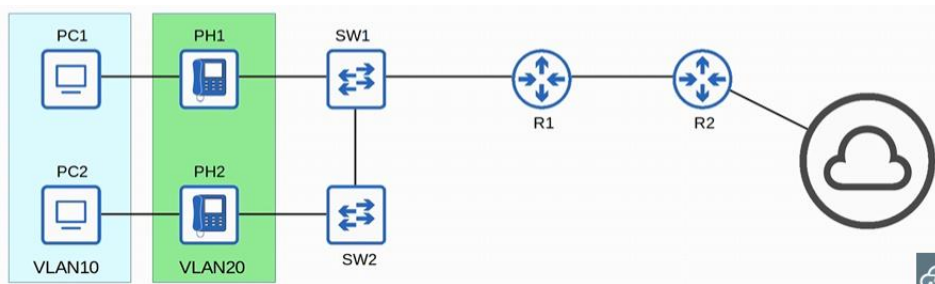
Because PCP exists only inside the **802.1Q VLAN tag**, it can only be used where Ethernet frames are tagged.

These include:

1. Trunk links
2. Access links configured with a Voice VLAN

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Network Topology

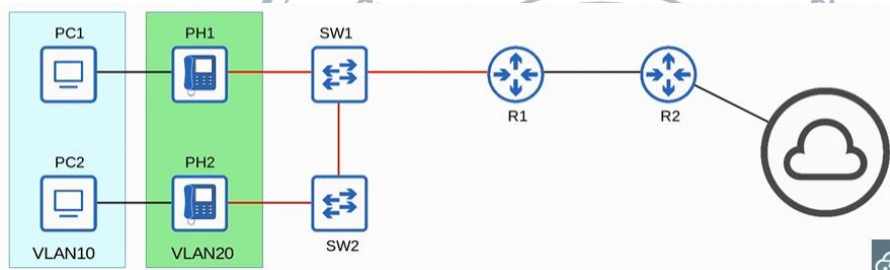


Links Carrying 802.1Q Tags

Assuming:

- the network is fully configured,
- all devices can communicate,
- the native VLAN is not being used,

the tagged links are:



Legend: - - - - - 802.1Q tagged links

Tagged connections are:

1. Phone ↔ Switch (Access port with Voice VLAN)
2. SW1 ↔ SW2 (Trunk)
3. SW1 ↔ R1 (Trunk)

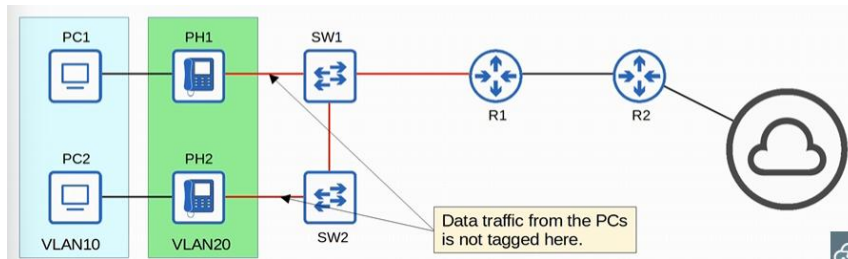
On these links:

- Devices can **mark** packets using PCP.
- Receiving devices use those markings to classify traffic as high, medium, low, or other priority levels.

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PCP Limitation

Traffic from the PCs is **not tagged** between the PC and IP phone.



Therefore, PCP cannot be used on these PC-to-phone links.

In addition:

1. Traffic between **R1 and R2**
2. Traffic between **R2 and external destinations**

also does **not** contain an 802.1Q tag.

Since there is no dot1q tag:

- No PCP field exists.
- Traffic cannot be marked with PCP.
- PCP cannot be used for classification.

This is the major limitation of Layer 2 PCP classification. Layer 3 classification methods (**such as DSCP**) do not have this limitation.

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Layer 3 QoS Classification

In the **IPv4 header**, there is a byte called the **Type of Service (ToS) byte**.

IPv6 also has a QoS field called the Traffic Class byte, but the focus here is on IPv4.

The ToS byte is the second byte of the IPv4 header, immediately after the Version and IHL fields.

IPv4 Header

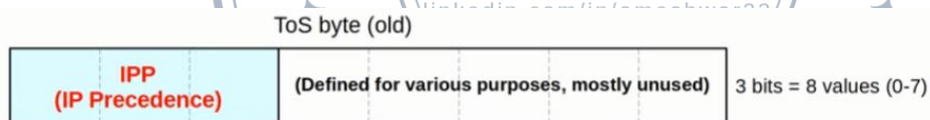
Offsets	Octet	0	1	2	3
Octet	Bit	0	1	2	3
0	0	Version	IHL	DSCP	ECN
4	32	Identification	Flags	Total Length	Fragment Offset
8	64	Time To Live	Protocol	Header Checksum	
12	96	Source IP Address			
16	128	Destination IP Address			
20	160				
24	192				
28	224				
32	256				

The modern ToS byte contains two fields:

1. **DSCP** (Differentiated Services Code Point)
2. **ECN** (Explicit Congestion Notification)

Previous Use of the ToS Byte (IPP)

Previously, the ToS byte had a different format:



The **IPP (IP Precedence)** field:

- Used 3 bits.
- Marked packets according to priority, similar to the PCP field in the 802.1Q header.
- The remaining 5 bits were defined for other purposes but were mostly unused.
- Three bits provide 8 possible values (0–7).

Current ToS Byte (DSCP)

The current ToS byte is organized differently:



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Changes include:

1. **6 bits** allocated to DSCP.
2. **2 bits** allocated to ECN.

Using 6 bits allows 64 possible DSCP values (0–63), providing much greater flexibility for marking and classifying traffic.

IP Precedence (IPP) Standard Markings

Before discussing DSCP in detail, the standard IPP markings are reviewed.

These markings are similar to PCP.

IPP Value	Purpose
7	Network control (reserved)
6	Network control (reserved)
5	Interactive voice traffic
4	Interactive video traffic
3	Voice signaling traffic (setting up and tearing down phone calls)
0	Best effort traffic (regular data traffic)

Since 6 and 7 are reserved for **network control traffic** (such as OSPF messages exchanged between routers), only six usable priority values remain.

Although six values are sufficient for many networks, more complex networks require greater flexibility.

Why DSCP Replaced IPP

Because:

- IPP used only **3 bits** of the ToS byte.
- The remaining **5 bits** were largely unused.

An additional **3 bits** were added to IPP, creating **DSCP (Differentiated Services Code Point)**.

This significantly increased the number of available markings.

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DSCP Standard

RFC 2474, published in 1998, defines the DSCP field.

Later **Differentiated Services (DiffServ) RFCs** further define how the field should be used.

1. After IPP became DSCP, new standard markings were established.
2. Having standardized markings provides several benefits:
3. Simplifies QoS design.
4. Simplifies QoS implementation.
5. Improves interoperability between ISPs and enterprise networks because both sides use agreed-upon markings.
6. Provides additional operational benefits.

Standard DSCP Markings

Several industry-standard DSCP marking groups were developed.

A. Default Forwarding (DF)

Used for:

1. Best effort traffic
2. Traffic with no special QoS requirements

B. Expedited Forwarding (EF)

Used for traffic requiring:

1. Low loss
2. Low latency
3. Low jitter

Typically used for **voice traffic**.

C. Assured Forwarding (AF)

Assured Forwarding is not a single marking.

It consists of 12 standard DSCP values.

D. Class Selector (CS)

Class Selector is a set of **8 standard values**.

Its purpose is to provide backward compatibility with IPP.

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QoS Configuration Example

QoS configuration is not part of the CCNA exam, but the following example shows how DSCP values appear during configuration.

CLI Output

```
R1(config)#class-map TEST
R1(config-cmap)#match dscp ?
<0-63>    Differentiated services codepoint value
af11      Match packets with AF11 dscp (001010)
af12      Match packets with AF12 dscp (001100)
af13      Match packets with AF13 dscp (001110)
af21      Match packets with AF21 dscp (010010)
af22      Match packets with AF22 dscp (010100)
af23      Match packets with AF23 dscp (010110)
af31      Match packets with AF31 dscp (011010)
af32      Match packets with AF32 dscp (011100)
af33      Match packets with AF33 dscp (011110)
af41      Match packets with AF41 dscp (100010)
af42      Match packets with AF42 dscp (100100)
af43      Match packets with AF43 dscp (100110)
cs1       Match packets with CS1(precedence 1) dscp (001000)
cs2       Match packets with CS2(precedence 2) dscp (010000)
cs3       Match packets with CS3(precedence 3) dscp (011000)
cs4       Match packets with CS4(precedence 4) dscp (100000)
cs5       Match packets with CS5(precedence 5) dscp (101000)
cs6       Match packets with CS6(precedence 6) dscp (110000)
cs7       Match packets with CS7(precedence 7) dscp (111000)
default   Match packets with default dscp (000000)
ef        Match packets with EF dscp (101110)
```

Explanation

The command: `match dscp ?`
displays all available DSCP matching options.

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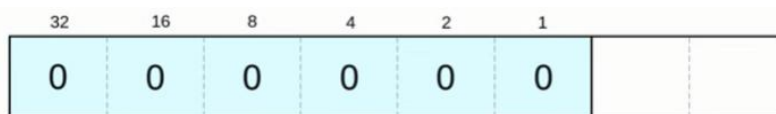
1. Default Forwarding (DF)

DF is used for **best-effort traffic**.

- Recommended DSCP value: 0
- All six DSCP bits are 0.

To make the binary values easier to understand, the decimal value of each binary position is shown above the bits.

DF DSCP Format



Regular network traffic, such as sending email or downloading files, typically uses:

1. **Binary:** 000000
2. **DSCP:** 0

This indicates that the traffic **does not require any special QoS treatment**.

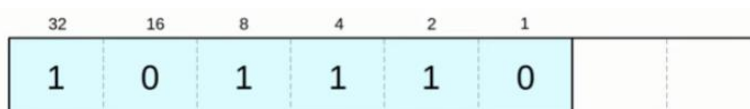
2. Expedited Forwarding (EF)

EF is used for traffic requiring: [linkedin.com/in/omeshwar23](https://www.linkedin.com/in/omeshwar23)

1. Low loss
2. Low latency
3. Low jitter

Voice traffic is typically marked as **EF**.

EF DSCP Format



1. **Binary:** 101110
2. **DSCP:** 46

Marking Purpose

DSCP

DF Best-effort traffic

0

EF Low-loss, low-latency, low-jitter traffic (typically voice)

46

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3. Assured Forwarding (AF)

After DF and EF, the next standard marking is **AF (Assured Forwarding)**.



af11	Match packets with AF11 dscp (001010)
af12	Match packets with AF12 dscp (001100)
af13	Match packets with AF13 dscp (001110)
af21	Match packets with AF21 dscp (010010)
af22	Match packets with AF22 dscp (010100)
af23	Match packets with AF23 dscp (010110)
af31	Match packets with AF31 dscp (011010)
af32	Match packets with AF32 dscp (011100)
af33	Match packets with AF33 dscp (011110)
af41	Match packets with AF41 dscp (100010)
af42	Match packets with AF42 dscp (100100)
af43	Match packets with AF43 dscp (100110)

AF is more complex because it defines:

- Four traffic classes
- Three drop precedence levels within each class

Traffic Classes

AF defines four traffic classes.

All packets in the **same class** have the **same priority**.

A higher-class number means:

- Higher priority
- Better forwarding service than lower classes

Drop Precedence

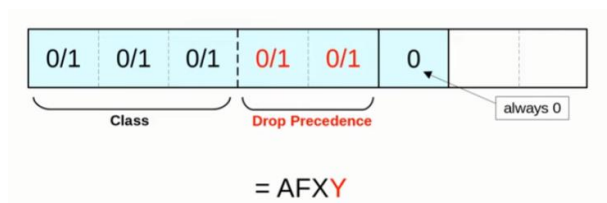
Within each class are three drop precedence levels.

A higher drop precedence means:

The packet is more likely to be **dropped during** congestion because of **WRED**.

AF Bit Layout

In AF, the six DSCP bits are interpreted as follows.



Format: AF X Y

Where:

1. Last bit is always 0

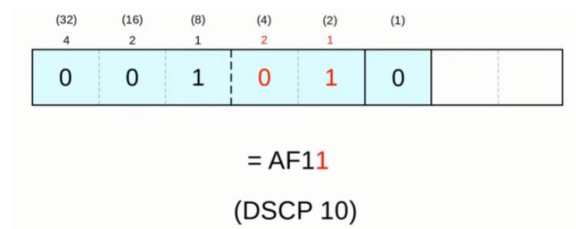
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2. First 3 bits = Class
3. Next 2 bits = Drop Precedence

The AF value is written as: AF X Y

AF Example – AF11

Binary DSCP: 001010

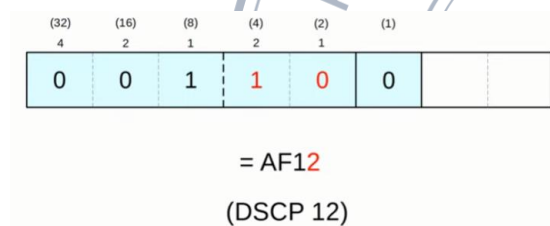


Result: AF11 = DSCP 10

AF is simply a standardized naming system built on normal 6-bit DSCP values.

AF Example – AF12

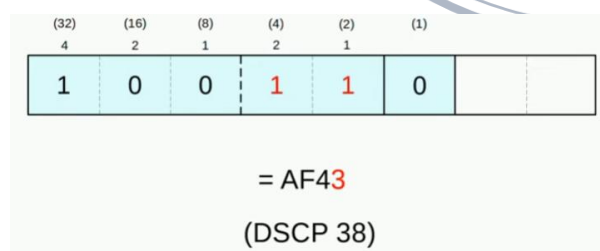
Binary: 001100



Result: AF12 = DSCP 12

AF Example – AF43

Binary: 100110



Result: AF43 = DSCP 38

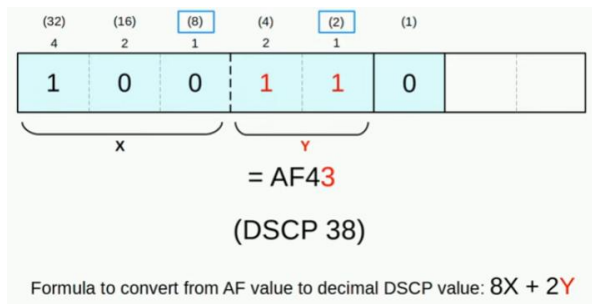
Within AF:

- AF43 is the highest AF value.
- There are no Class 5, Class 6, or Class 7 AF values.

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AF to DSCP Conversion Formula

Instead of converting from binary every time, the following shortcut can be used:



$$\text{DSCP} = (8 \times X) + (2 \times Y)$$

Where: X = Class, Y = Drop precedence

Reason:

1. **8** is the lowest decimal value represented by the class portion.
2. **2** is the lowest decimal value represented by the drop precedence portion.

Although this shortcut is useful, it is still important to understand the underlying binary representation.

Complete AF Summary

	Lowest drop precedence		Highest drop precedence
Highest priority	AF4 1 (34)	AF4 2 (36)	AF4 3 (38)
	AF3 1 (26)	AF3 2 (28)	AF3 3 (30)
	AF2 1 (18)	AF2 2 (20)	AF2 3 (22)
Lowest priority	AF1 1 (10)	AF1 2 (12)	AF1 3 (14)

↑ Higher class = Higher priority

→ Higher drop precedence = More likely to be dropped

Therefore:

1. **AF41** receives the **best treatment** because it is in the highest priority class with the lowest drop precedence.
2. **AF13** receives the **worst treatment** because it is in the lowest priority class with the highest drop precedence.

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4. Class Selector (CS)

CS (Class Selector) is the simplest DSCP marking group.

cs1	Match packets with CS1 (precedence 1)	dscp (001000)
cs2	Match packets with CS2 (precedence 2)	dscp (010000)
cs3	Match packets with CS3 (precedence 3)	dscp (011000)
cs4	Match packets with CS4 (precedence 4)	dscp (100000)
cs5	Match packets with CS5 (precedence 5)	dscp (101000)
cs6	Match packets with CS6 (precedence 6)	dscp (110000)
cs7	Match packets with CS7 (precedence 7)	dscp (111000)

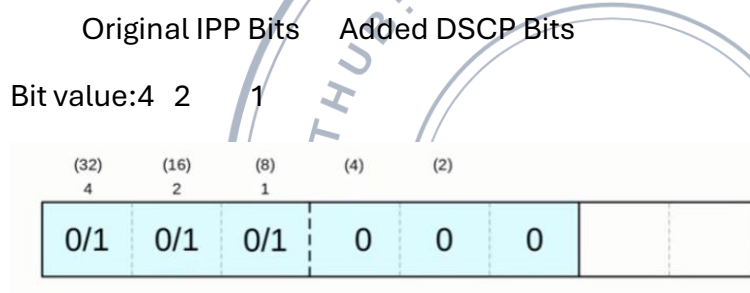
Its purpose is to provide **backward compatibility** with IPP (IP Precedence).

How Backward Compatibility Works

DSCP uses 6 bits, but for CS:

1. The original 3 IPP bits are kept.
2. The 3 additional DSCP bits are always 0.

CS Bit Format



The three left bits represent the original **IPP values**, giving **8 possible values (0–7)**.

	(32) 4	(16) 2	(8) 1	(4)	(2)		
	0/1	0/1	0/1	0	0	0	
IPP:	0	1	2	3	4	5	6 7
CS:	CS0	CS1	CS2	CS3	CS4	CS5	CS6 CS7
DSCP: (decimal)	0	8	16	24	32	40	48 56

CS to Decimal DSCP Conversion

Although CS is a standardized DSCP naming scheme, each CS value is still a normal DSCP value.

Conversion is simple:

$$\text{DSCP} = 8 \times \text{CS Number}$$

Resulting values: 0 8 16

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Standard Use of DSCP Values

At this point the standard marking groups have been introduced: DF, EF, AF, CS

The next question is how these values should actually be used.

RFC 4954, developed with the help of Cisco, brings these markings together and provides standardized recommendations.

Although the RFC contains many recommendations, the key ones are:

Traffic Type	Recommended Marking
Voice traffic	EF
Interactive video	AF4x (AF41, AF42, AF43)
Streaming video	AF3x
High-priority data	AF2x
Best-effort data	DF (DSCP 0)

These are **standard recommendations**.

The network engineer designing the QoS policy ultimately decides which traffic receives which markings.

Trust Boundaries

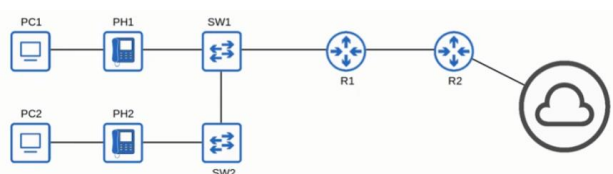
A trust boundary defines where network **devices trust or do not trust** the QoS markings on received packets.

Behaviour Inside and Outside the Trust Boundary

Trusted markings: The device forwards the packet without changing the QoS markings.

Untrusted markings: The device changes the markings according to the configured QoS policy before forwarding the packet.

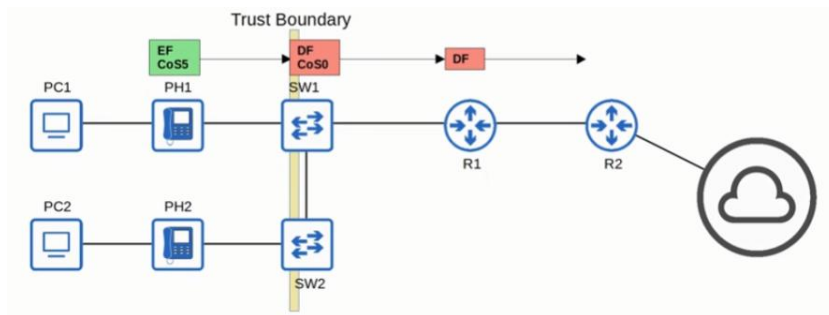
Network Topology



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Trust Boundary at SW1

Assume the trust boundary is located at **SW1**.



Phone1 sends a packet marked: **EF, CoS 5**

CoS refers to the PCP field in the 802.1Q header.

The terms CoS5 and PCP5 refer to the same marking.

Since Phone1 is **outside the trust boundary**, SW1 does not trust those markings.

According to the configured policy, SW1 may:

- Change DSCP EF to DF.
- Change CoS 5 to CoS 0.

The packet is then forwarded to R1, then R2.

Since there is no 802.1Q header beyond the switch, only the **DSCP (DF)** marking remains.

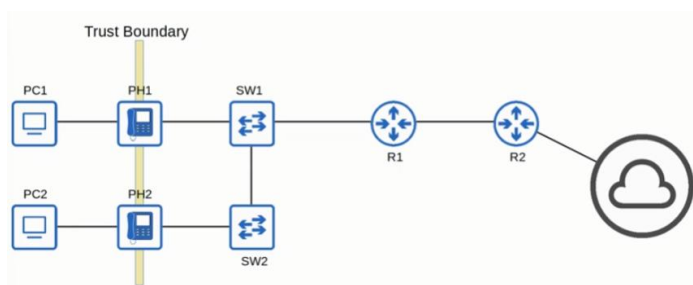
Recommended Trust Boundary

The previous configuration is **not ideal**.

Normally, traffic from an IP phone should be trusted, because voice traffic should receive high priority.

If an IP phone is connected to the switchport, the recommended practice is to move the trust boundary to the IP phone.

This is configured on the switch port connected to the IP phone, not on the phone itself.



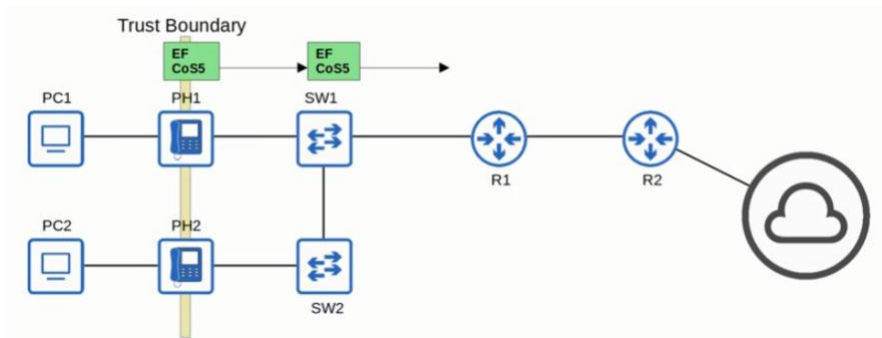
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Trusted Phone Traffic

With the trust boundary moved to the IP phone:

When Phone1 sends a packet marked: EF, CoS5

SW1 trusts those markings and forwards the packet without changing them.



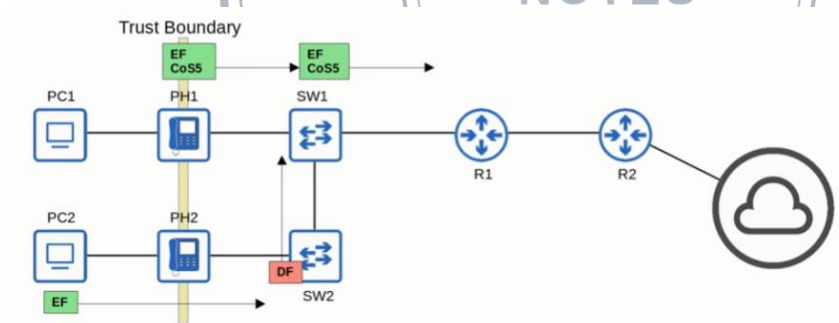
Untrusted PC Traffic

Now consider **PC2**.

If PC2 sends a packet marked **EF**, the switch **should not trust** that marking.

For example, SW1 may change: **EF (DSCP 46) → DF (DSCP 0)**

This prevents ordinary PC applications from receiving the same priority as voice traffic.



Queuing/Congestion Management

Why Queuing/Congestion Management Is Important?

When a network device receives traffic **faster than it can forward it** through the appropriate outgoing interface, packets are temporarily stored in that interface's **queue**.

If the queue becomes full:

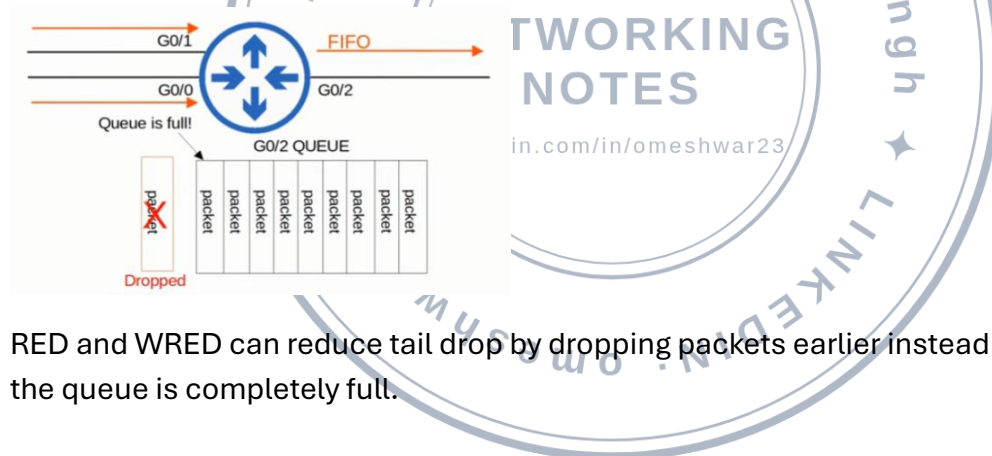
- New packets that cannot fit into the queue are discarded.
- This is called **tail drop**.

Previously introduced **RED** and **WRED** help prevent tail drop by dropping packets before the queue becomes completely full.

Single Queue (FIFO)

Previously, packets were forwarded using a FIFO (First In, First Out) queue.

When the queue fills up, additional packets are dropped.



RED and WRED can reduce tail drop by dropping packets earlier instead of waiting until the queue is completely full.

Multiple Queues

An important part of QoS is using **multiple queues**, not just a single queue.

This is where **traffic classification** becomes important.

A device can classify traffic using various factors, such as:

- DSCP value in the IP header
- Other classification methods

After classification, packets are placed into the appropriate queue.

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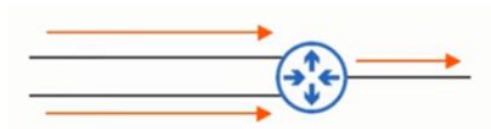
Although multiple queues exist, only one frame can be transmitted from an interface at a time.

A scheduler determines which queue is served next.

Prioritization allows some queues to receive higher priority than others.

Multiple Queue Operation:

In this example, the router has four output queues.



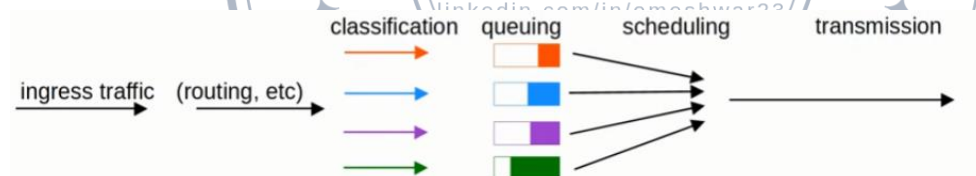
Packets are classified (for example, by their DSCP markings) and placed into the appropriate queue.

The scheduler determines:

- Which queue to transmit from
- How much traffic to send
- The transmission order

Packets are then transmitted **one packet at a time**.

This diagram is a simplified representation of the process.



After the router determines the outgoing interface, the packet passes through:

1. Classification
2. Queuing
3. Scheduling
4. Transmission

Weighted Round Robin (WRR):

A common scheduling method is Weighted Round Robin.

Round Robin means:

- The scheduler services each queue **in order**.
- The process repeats **cyclically**.

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Weighted means:

- Higher-priority queues transmit more data each time the scheduler visits them.
- Higher-priority queues therefore receive greater output.

CBWFQ (Class-Based Weighted Fair Queuing):

CBWFQ is a commonly used scheduling method.

It combines:

- Weighted Round Robin scheduling
- Guaranteed minimum bandwidth for each queue during congestion.

CBWFQ Operation

Traffic is:

1. Classified.
2. Placed into queues.
3. Scheduled using Weighted Round Robin.
4. Transmitted.

Each queue receives:

- A guaranteed percentage of the interface bandwidth.
- This guarantee applies even during congestion.

Example:



This approach is significantly more advanced than using a single output queue.

Limitation of Weighted Round Robin

Although CBWFQ guarantees bandwidth, it is **not ideal for voice and video traffic**.

Even high-priority queues must wait for their turn in the scheduler.

This waiting can introduce:

- Delay
- Jitter

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LLQ (Low Latency Queuing):

To reduce delay and jitter, LLQ (Low Latency Queuing) can be configured.

LLQ designates **one or more** queues as **strict priority** queues.

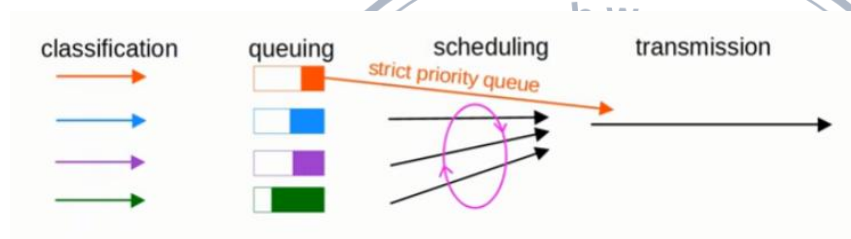
Strict Priority Queue

A strict priority queue works as follows:

- If the queue contains traffic
- The scheduler always selects packets from that queue first.
- The scheduler continues serving that queue until it becomes empty.

This greatly reduces: Delay & Jitter

for high-priority traffic such as voice and video.



As soon as packets enter the priority queue, the scheduler forwards them immediately.

Limitation of LLQ:

LLQ has one important drawback.

If the strict priority queue **always contains traffic**, it can consume all available transmission opportunities.

As a result:

- Other queues may never transmit.
- Those queues can become **starved**.

Preventing Queue Starvation

Policing can control the amount of traffic allowed into the strict priority queue.

This prevents the priority queue from consuming all of the interface bandwidth.

Shaping/Policing

What Is Traffic Shaping/Policing?

At the CCNA level, the main objective is to understand what these two QoS functions do.

Both traffic **shaping** and traffic **policing** are used to **control the rate of traffic**.

In the previous queuing examples, interfaces were assumed to be operating:

At full capacity, or beyond full capacity, which required packets to be queued.

However, there are situations where it is desirable to **limit the traffic** rate below the physical capacity of the link.

Traffic Shaping:

Traffic shaping buffers traffic in a queue whenever the traffic rate exceeds the configured rate.

The concept is the same as queuing:

- Instead of the physical link capacity limiting traffic,
- A configured maximum traffic rate becomes the limiting factor.

Traffic exceeding the configured rate is **queued** instead of being sent immediately.

Traffic Policing:

Traffic policing is less forgiving than shaping.

If the traffic rate exceeds the configured rate: Excess traffic is **dropped**.

However, policing allows some flexibility.

A limited amount of **burst traffic** above the configured rate is allowed for a short period.

This accommodates applications that naturally send data in bursts instead of as a continuous stream.

Both of the following are configurable:

- The policed data rate
- The amount of burst traffic allowed

Computer Networks - CCNA

Classification with Shaping and Policing

Just like other QoS features, **classification** can be used with shaping and policing.

Different traffic classes can be assigned **different traffic rate limits**.

Network Topology



Shaping and Policing in the Example

In this example:

- The Customer Router performs **outbound traffic shaping** on its **G0/0** interface.
- The **ISP Router** performs **inbound traffic policing** on its **G0/0** interface.



Why Use Shaping and Policing?

Assume:

- Both physical interfaces are **Gigabit Ethernet (1000 Mbps)**.
- The customer has purchased only a **300 Mbps** Internet connection.

The ISP's policy is: "You paid for a 300 Mbps connection, so incoming traffic will be policed to 300 Mbps."

The customer realizes: "If traffic is sent faster than 300 Mbps, the ISP will drop the excess traffic."

To avoid those drops, the customer configures:

- Outbound shaping
- Traffic rate limited to 300 Mbps

This keeps the transmitted traffic within the contracted rate.